

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) | (i)   |  | award (1) for correct order<br><br>zinc carbonate<br>copper(II) carbonate<br>lead carbonate<br><br>accept zinc / Zn<br>accept copper / Cu<br>accept lead / Pb                                                                                                                                                           |                 |     | 1   | 1     |       | 1    |
|          |     | (ii)  |  | Sodium carbonate only decomposes a small amount on heating <input type="checkbox"/><br><br>Sodium carbonate is very unstable <input type="checkbox"/><br><br>Sodium carbonate does not decompose on heating <input checked="" type="checkbox"/><br><br>Sodium carbonate decomposes too quickly <input type="checkbox"/> |                 |     | 1   | 1     |       | 1    |
|          |     | (iii) |  | 70 (2)<br><br>if incorrect award (1) for $\frac{3.5}{5.0} / 0.7$                                                                                                                                                                                                                                                        |                 | 2   |     | 2     | 2     |      |
|          |     | (iv)  |  | Cu <sup>2+</sup>                                                                                                                                                                                                                                                                                                        |                 | 1   |     | 1     |       |      |
|          | (b) |       |  | yellow / orange / yellow-orange                                                                                                                                                                                                                                                                                         | 1               |     |     | 1     |       | 1    |
|          |     |       |  | Question 6 total                                                                                                                                                                                                                                                                                                        | 1               | 3   | 2   | 6     | 2     | 3    |